



From Regional Strengths to National Impact

A Place-Based STI Strategy for the Dutch Alternative Protein Ecosystem

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Policy Brief





Introduction

The Netherlands has formally established the “protein transition” as a national strategic priority, essential for achieving sustainable economic growth and aligning with UN’s SDGs. This ambition is to reach 50% of alternative protein in public diet consumption in 2030, backed up by the National Protein Strategy (*Nationale Eiwitstrategie*) (Ministerie van Landbouw, Natuur en Voedselkwaliteit, 2020). To fund this transition, the Netherlands is making significant investments, with the National Growth Fund (*Nationaal Groeifonds*) allocating hundreds of millions of euros to proposals designed to secure a leading international position in cellular agriculture and alternative proteins (Rijksoverheid, 2022).

However, despite clear national commitment and world-class R&D, the Netherlands faces a critical “valley of death” (Podder et al., 2022). The translation of scientific leadership into national economic impact is failing. A 2024 joint report by Foodvalley NL and Invest-NL identifies significant obstacles in scaling up for innovative companies. These obstacles include “limited access to facilities, high operational costs, complex regulations, and funding shortages”, which collectively create a severe bottleneck in the innovation pipeline (Foodvalley NL & Invest-NL, 2024, p. 2).

This policy brief argues that the scale-up failure is a direct consequence of a structural fragmentation within national science technology and innovation (STI) ecosystem. This system is currently separated into two highly specialized but poorly integrated provincial hubs.

Gelderland Province operates as the national deep tech R&D engine. It has great records in generating foundational patents and early-stage research. Meanwhile, Noord-Holland Province functions as the capital and market hub centered in Amsterdam (Rendel, 2025).

This disconnection between the two provinces has been making publicly-funded R&D fails to find the necessary domestic scale-up capital and commercial expertise. This brief provides an evidence-based analysis of this potential regional gap and proposes a targeted, place-based STI strategy to bridge it, ensuring that the Netherlands’ strategic public investments translate into national economic impact.

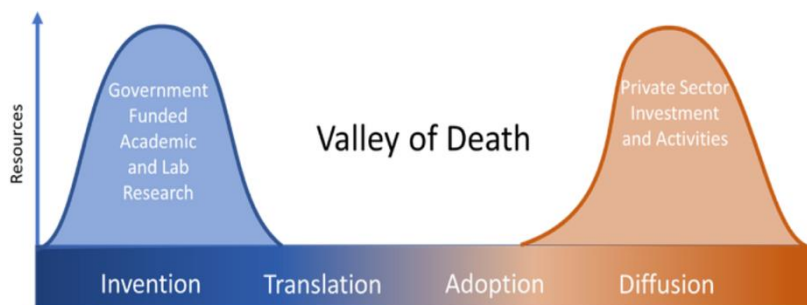


Figure 1 The “Valley of Death” is a metaphor for how long it can take to scale up innovations and inventions. (Podder et al., 2022)

The Comparative Regional Innovation System (CRIS) Approach

The findings of this brief are based on a Comparative Regional Innovation System (CRIS) framework. It examines the functional differences and further potential collaborations between Gelderland and Noord-Holland. This framework emphasizes the interplay between different actors which includes institutions, firms and government, and how their interactions shape the flow of knowledge from innovation to market application.

1. Publication and patent analysis

Calculated and compared the number and trends of scientific and technological outputs in these two regions by using the data from 2010 to 2024.



2. Network analysis

Constructed collaboration networks based on institutional affiliation in publications and patent co-applicants. The degree centrality was used to quantify the connections among actors across regions, measuring overall levels of system connectivity and knowledge exchange.

3. Actor mapping and integration

Visualize and integrate previous analyses across knowledge production, technological outputs, market activation (firm types) of protein transition.

4. Policy analysis

A qualitative review of regional and national policy documents was conducted to examine whether current interventions primarily reinforce regional specialization or promote coordination across the protein transition value chain.

Table 1 Indicators used in the analysis and the reasons for choosing them.

	Indicator	Data	Reason
Key indicators	Publication Number	Scopus	Evaluate their scientific innovation capabilities.
	Patent Number	Lens.org	Evaluate their technological innovation capabilities.
	Actors Mapping	Food Valley	Analyze the roles of relevant participants in each region (such as research institutions, suppliers, producers, service providers, etc.).
Supplementary indicators	Patent Network	Lens.org	Reveal the level and structures of collaboration within and between regions.
	Policy Review	Government websites	Identify whether local policy support prioritizes R&D, production, or market commercialization.

Analysis: Evidence of a “Two-Silo” Ecosystem

Evidence of the R&D-Technology Mismatch

Currently, Noord-Holland has shown a relatively strong advantage for technological and industrial transformation, particularly in Amsterdam, which has become an important hub for alternative protein and food-tech startups and incubators (Impact Hub Amsterdam, 2024). Meanwhile, Gelderland, led by WUR’s knowledge production, continues to play a key role in national research and academic output on the protein transition. WUR’s ongoing projects and retail-chain monitoring initiatives such as Monitor Protein Transition 2023 that emphasizes systemic change rather than immediate technological output.

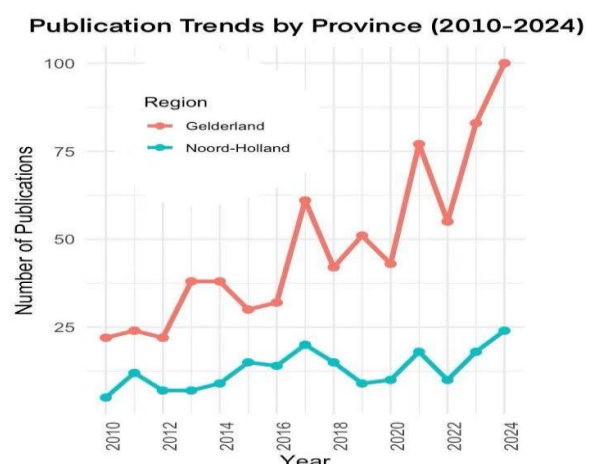


Figure 2 Comparative Publication Trends Between Gelderland and Noord-Holland (2010–2024)

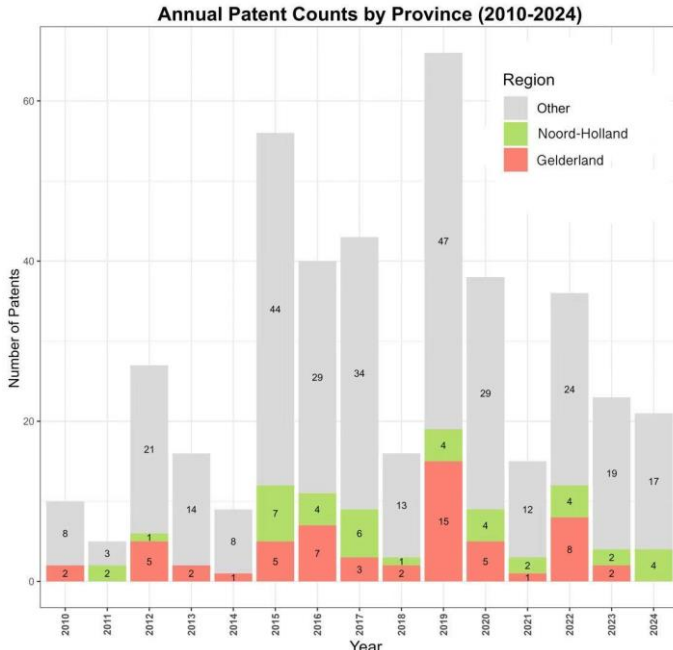


Figure 3 Regional Distribution of Patent Production in the Netherlands (2010–2024)

As observed, the publications in Gelderland are dramatically increasing and reach to 100 in 2024 most probably led by WUR’s knowledge production (Figure 2), whereas this research advantages fail to into consistent local technological success, as evidence by its patent number drop to zero in the same year.

However, Noord-Holland has emerged as the practical engine for technological innovation, producing nearly a quarter of all patents in 2024. Despite this, its capacity of knowledge creation remains limited, with annual publications fluctuating around 20. Thus, regions like Gelderland may experience rapid growth in research and publications, but if local mechanisms for technology transfer, industrial chain connectivity, patent incentives, and investment environments remain inadequate, they may risk exhibiting a pattern of “abundant research but limited practical use.”

Network Analysis Reveals a Critical R&D-Market Disconnect

Based on Dutch patent applicant collaboration network, it is concerning that core innovation activities are almost concentrated within Gelderland. The network shows several small, self-contained subnetworks, dominated by research-oriented actors such as WUR and nearby industrial partners. Besides, there is a notable absence of connection between these clusters, indicating a limited degree of inter-regional collaboration.

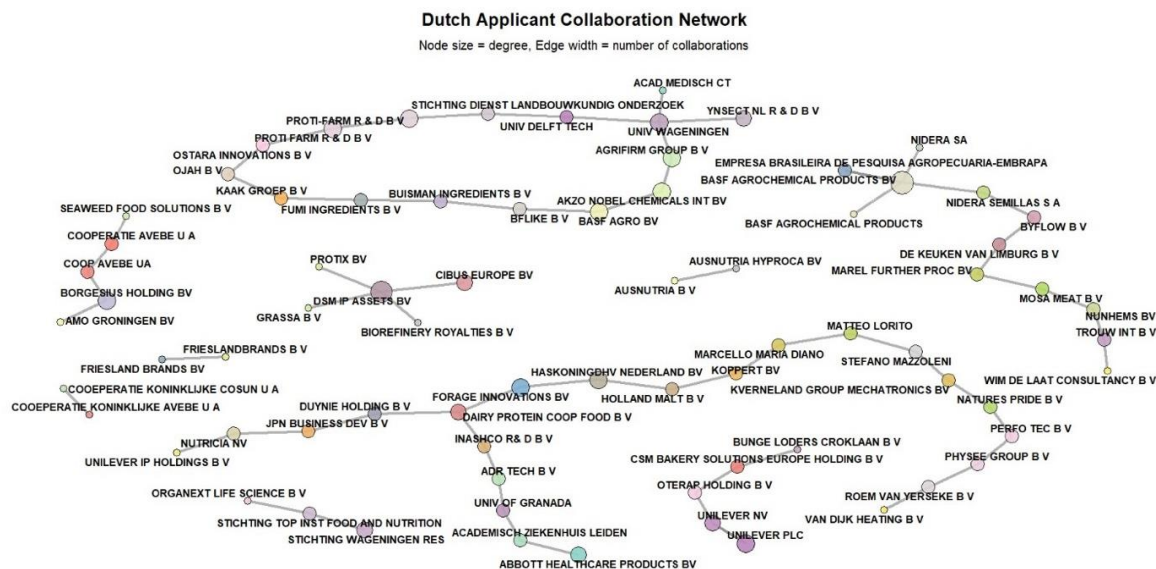


Figure 4 Collaboration Network of Patent Applicants in the Netherlands



Furthermore, the Noord-Holland region, and particularly Amsterdam, which is an important center for market activity, distribution channels and end-user connections, is not a significant player in this collaboration network. This absence indicates that market-driven actors are not yet fully integrated into research-driven innovation ecosystems.

Functional Divides Reveal a Path to Synergy

The functions between two regions have obvious divisions based on the number of actors within different sectors, publications. Noord-Holland has strong commercial, investment, and market-oriented capabilities, including product distribution, market visibility, and commercialization potential, even though its scientific and technical output remains limited compared to Gelderland, which is more specialized in innovation-related generation, particularly in research output and service provision. Thus, this complementary highlight the potential for interregional collaboration, where Gelderland can provide advanced innovation and Noord-Holland can enhance dissemination, adoption, and market impact.

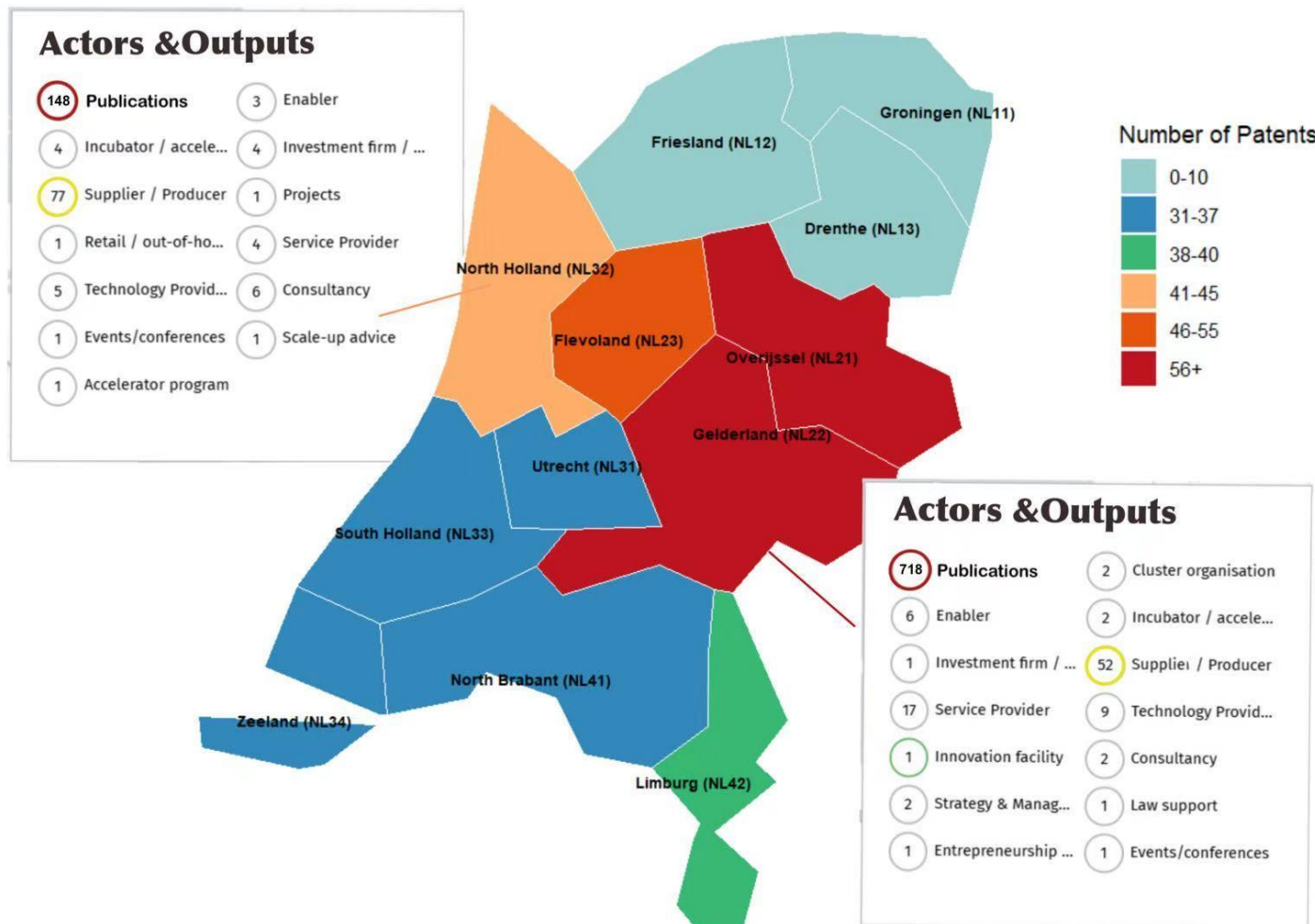


Figure 5 Provincial Patterns of Publication & Patent Intensity and Innovation Actors of Alternative Protein in the Netherlands

The Need for a Strategic Platform and Talent Mobility

In the Netherlands, the protein transition is facing a huge challenge: its world leading research and market capabilities operate in silos, making it difficult to form a unified innovation ecosystem.

The key issue is the lack of a coordinated and assessed system, which is a major obstacle to the development of an innovation ecosystem that can support itself. In the current system, market application efficiency is rarely weighted in innovation assessment, and market demand fails to totally shape research directions, leading to a gap between innovation and real-world diffusion.

Thus, it is hard to make full use of its own innovations due to limited internal knowledge transfer and practical application. Besides, the absence of a strategic and integrative platform, with systematic evaluation mechanisms and tools to identify complementarities, has resulted in fragmented policy interventions rather than comprehensive and systematic strategies.

This gap in systems and data is reinforced at the human level. Gelderland has a concentration of specialized higher education talent, including researchers, PhDs, and food scientists. By contrast, Noord-Holland is defined by its highly international ecosystem and diverse workforce of industry professionals, including venture capitalists, brand managers, and global marketing experts. These two talent pools operate in different worlds. Bridging this human gap through initiatives like placing professionals in both a WUR lab and an Amsterdam accelerator is therefore useful to building a hybrid talent base that can successfully lead the transition from invention to market.

Weighing the Options: The Imperative for a Coordinated Bridge

Given the evidence of the regional fragmentation, policymakers must consider alternative strategies for intervention. This brief will present two primary pathways: (1) a regional specialization approach that relies on market forces and existing regional strengths, and (2) a coordinated ecosystem bridge approach that actively links them.

The Regional Specialization Approach

This strategy would continue to support Gelderland as a world-class R&D provider and Noord-Holland as a premier capital and market hub. In this model, collaboration is left to organic, market-driven interactions. This approach respects market dynamics, fosters competition, and allows each region to maximize its core competency.

However, this strategy fails to address the central challenge identified in our analysis, the “scale-up valley of death”. This oversight creates two significant national risks.

- (i) **Innovation Leakage:** Relying on the market ignores the “European Paradox”, where leading R&D in Gelderland fails to translate into commercial leadership (Audretsch, 2007). Without a funded domestic path, Gelderland’s deep tech startups which are seeking high-risk scale-up capital will likely be acquired by foreign investors. This is a known risk in the Dutch food sector, where R&D often leaks and is commercialized abroad (Rabobank, 2023).
- (ii) **Market Shallowness:** Meanwhile, Noord-Holland’s capital lacks access to new R&D outcomes and will likely concentrate on shallow market innovations like first-generation plant-based products. These risks missing the high value deep tech frontier such as precision fermentation, or cellular agriculture, and industry experts identify them as the next major growth field (GFI, 2024).

The Coordinated Ecosystem Bridge Approach

This strategy involves a policy intervention and wants to build a formal link between the two regions, this is consistent with modern place-based innovation policies (Foray, 2018).

This is the strategy directly addresses the identified “scale-up gap”. By co-funding shared pilot and production facilities, it lowers the risks of investment for Noord-Holland’s capital and provides a clear commercialization path for Gelderland’s R&D.

This model aligns with international best practices: **Singapore** actively funds open-access R&D and pilot-scale facilities through its A*STAR agency, specifically to attract global venture capital and anchor high-tech manufacturing (EDB Singapore, 2023). Moreover, **Denmark** Successfully operates “Food & Bio Cluster Denmark” as a single national platform connecting all universities and companies to ensure the entire value chain is cohesive (Food & Bio Cluster Denmark, 2024).

Despite the advantages, this approach is more complex, requiring more political will to overcome regional competition and secure cross-provincial agreements. It also carries the risk of creating a new bureaucracy if the coordinating platform is not designed to be flexible and industry focused.

Conclusion

While the Netherlands has a strong advantage in both research capabilities and dynamic market innovation in the protein transition sector, these strengths are not sufficiently connected. The country should focus on creating stronger regional networks, coordinating knowledge diffusion, and integrating evaluation frameworks into a coherent, self-reinforcing ecosystem. These approaches would help accelerate the practical adoption of sustainable protein solutions.

Actionable Policy Recommendations

1. Establish a **Netherlands Protein Innovation Platform** to serve as a national coordinating body that integrates fragmented data and strategically connects Gelderland’s tech R&D with Noord-Holland’s capital.
2. Establish a dedicated **Cross-Provincial Scale-Up Fund** that utilizes public-private co-investment to finance shared pilot facilities, with funding made conditional on formal collaboration between Gelderland's technology providers and Noord-Holland's capital investors.
3. Implement a **Talent and Knowledge Mobility Initiative** through joint appointments and cross-provincial PhD programs to accelerate knowledge transfer and break down cultural barriers between the two regional ecosystems.



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AI Statement

For this document, I used artificial intelligence (AI) as a tool to assist in improving the clarity and flow of the original text. Specifically, I employed OpenAI's ChatGPT to help with the following:

- **Text Polishing:** I provided the original draft of the text and asked for suggestions to improve its structure, wording, and readability.
- **Data Interpretation Support:** I asked ChatGPT to help me code with raw data and articulate possible explanations for the observed data. The tool provided alternative perspectives and wording that I then critically assessed and integrated into my own interpretation.
- **Generating Alternative Phrasings:** I used the AI to explore different ways of articulating complex concepts (e.g., "commercialization valley of death," "innovation silos") to enhance the impact and readability of the text.

Prompts used:

1. "Can you polish this text in more policy brief style?" - This prompt was used to request AI assistance in refining the original text I wrote.
2. I used AI to discuss and refine my understanding of patent networks and actors map, helping to structure the narrative that connects the quantitative findings.
3. I had a brainstorm with AI and used it to think of possible underlying problems and challenges in current situations of protein transition.

The final output reflects my understanding of the original content and learning goals, and I am fully responsible for the final submission.

End of AI statement